Unit 3: Electric Potential

Lesson 3.1

Electric Potential Energy in a Uniform Electric Field

Contents

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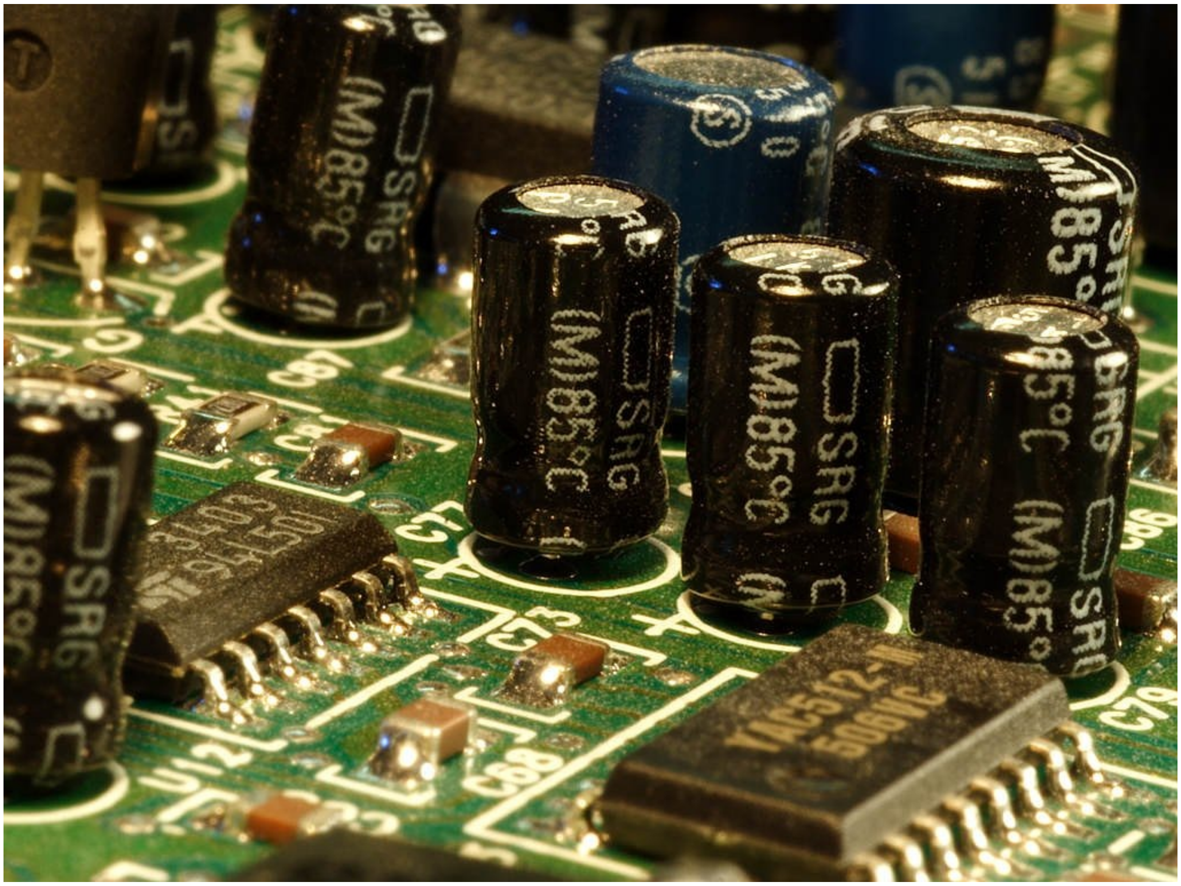

Introduction

To grasp the concept of electric potential energy, it is essential that we first understand energy as a whole. Energy, whether it is mechanical, chemical, or electrical, comes in two forms: kinetic and potential. Potential energy, which is the stored energy of an object at rest, is converted into kinetic energy when an object moves. All electrical circuits require the storage of energy, such as in the form of capacitors, in order for it to be transferred into light, heat, and motion, among many other forms. This stored energy in a circuit is what is referred to as electric potential energy. In this lesson, we will learn about the electric potential energy in a uniform electric field.

Learning Objectives

In this lesson, you should be able to do the following:

- Explain electric potential energy.
- Determine the electric potential energy in a uniform electric field.
- Calculate the electric potential energy of a system of charges.

DepEd Competency

Solve problems involving electric potential energy and electric potentials in contexts such as, but not limited to, electron guns in CRT TV picture tubes, conditions for merging of charge liquid drops, and Van de Graaff generators **(STEM_GP12EM-IIIc-22).**

Warm Up

Potential Energy in a Uniform Field

5 minutes

The simulation you are about to interact with will allow you to visualize how potential energy changes in a uniform field.

Materials

- potential energy in a uniform field simulator
- worksheet

Procedure

1. Set the simulator to its initial conditions. A screenshot of the simulator is presented in **Fig. 3.1.1**.

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[Potential Energy in a Uniform Field](http://physics.bu.edu/~duffy/semester2/c05_Uuniform.html)
Andrew Duffy, "Potential Energy in a Uniform Field," http://physics.bu.edu/~duffy/semester2/c05_Uuniform.html, last accessed on March 03, 2020

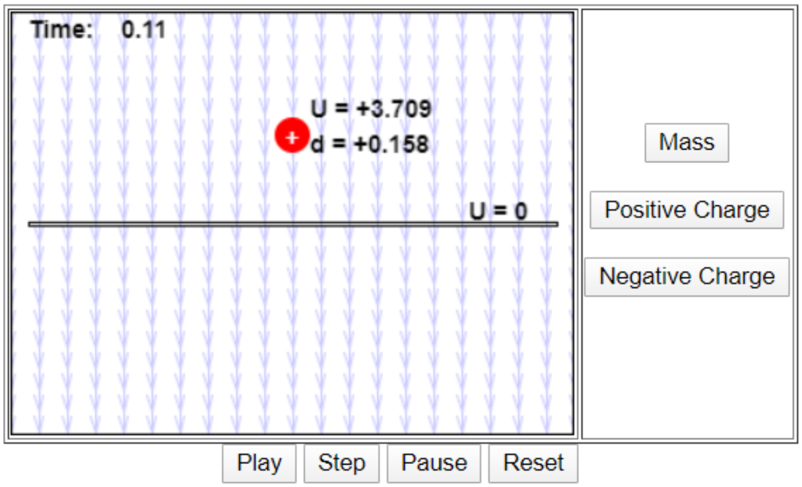

Fig. 3.1.1. Potential energy simulator

2. Set the simulator to positive charge, with the following values: U = +7.056 and d = +0.3. Play the simulation and pause it to record the potential energy and distance at each point in time provided in **Table 3.1.1**.
3. Set the simulator to negative charge, with the following values: U = -7.056 and d = -0.3. Play the simulation and pause it to record the potential energy and distance at each point in time provided in **Table 3.1.1**.

Data Table

Table 3.1.1. Electric potential energy in a uniform field

Charge	Time	Potential Energy (U)	Distance (d)
Positive	0		
	0.11		
	0.21		

Negative	0		
	0.11		

Guide Questions

1. What is the direction of the electric field in the simulation?
2. What is the direction of the positive charge? How about the negative charge?
3. What do the data you have gathered reveal about the relationship between the charge and its direction in the field? How about between the charge and the corresponding potential energy?

Learn about It!

What is electric potential energy?

Electric Potential Energy

Electric potential energy is most commonly compared with gravitational potential energy. Consider the situation in **Fig. 3.1.2**.

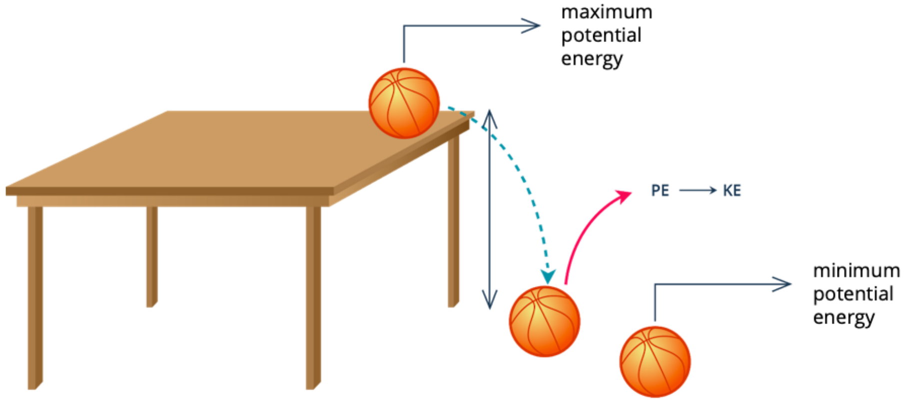

Fig. 3.1.2. Maximum and minimum gravitational potential energy

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A ball at rest situated on top of the table contains a large amount of stored, or gravitational potential, energy. Once it begins to fall off from the table, it accelerates and the potential energy is transformed into kinetic energy, until it reaches the floor and goes back to being at rest, where it possesses its minimum gravitational potential energy.

Electric charges in an electric field are comparable to the mass of objects in a gravitational field. The electric potential energy determines how much stored energy a charge has. Similar to the ball at rest on top of the table, a positive charge which is close to another positive charge contains high potential energy, and will be repelled when it moves. Placed in close proximity to a negative charge, however, it will have low potential energy, just like the ball after having fallen on the floor. **Electric potential energy** is thus defined as the energy needed to move a charge against an electric field.

Did You Know?

The term "potential energy" was first introduced by the Scottish physicist William Rankine in the 19th century.

Remember

Potential energy can only be defined for conservative forces, where the force required to move an object from one position to another is independent of the path taken. Electrostatic force, which is the force required in the interaction of charges, is an example of a conservative force.

• • •

Electric potential, or the amount of work required to bring the unit positive charge from infinity to a certain point, and electric potential energy, or the amount of energy required to move a charge against the electric field are two and separate terminologies.

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The General Equation

Electric force (F) is required to move an electric charge that contains potential energy either towards an opposite charge or against a like charge. The work done by this force may be expressed in terms of potential energy (U_E). When a charged particle moves from one point to another (from U_A to U_B), the change in its potential energy is mathematically expressed as follows:

$$W_{A\to B} = U_A - U_B = -(U_B - U_A) = -\Delta U$$

When the work done by the electrostatic force $W_{A\to B}$ is positive, U_A has more potential energy than U_B, and ΔU is negative as the potential energy decreases. During this displacement, the **change in the particle's kinetic energy** $\Delta K = K_B - K_A$ is equivalent to the **total work done** on the charged particle. Hence, the law of conservation of mechanical energy is also applicable to electric charges, and is mathematically expressed through the following:

$$K_A + U_A = K_B + U_B$$

Use **Table 3.1.2** as a reference for the names, symbols, and units of the quantities related to electric potential energy.

Calculating Electric Potential Energy

Recall that in a uniform electric field, the electric field lines are equally distributed parallel to each other, and therefore, the strength of the field does not change. The potential difference between two separate conducting plates create these uniform fields. Here, the work done by the electric field is given by the equation below:

$$W_{A\to B} = Fd = q_0Ed$$

where $W_{A\to B}$ is the work done by the force, F is the force, d is the displacement, and q_0E is the enclosed electric charge.

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Table 3.1.2. Quantities related to electric potential energy

	Symbol	Unit	
Work done by force	$W_{A \to B}$	J	joule
Electric potential energy	U_E	J or Nm	joule or newton meter
Force	F	N	newton
Enclosed electric charge	q_0E	C	coulomb
Radius	r	m	meter
Coulomb's constant	k	$9.0 \times 10^9 \frac{Nm^2}{C^2}$	newton square meter per square coulomb

Electric Potential Energy in a Uniform Field

Since the force is in a similar direction with the net displacement of the charge, as shown in **Fig. 3.1.3.**, the work is considered positive and the potential energy for the electric force is calculated as follows:

$$U_E = q_0 E d$$

where U_E is the electric potential energy, q_0E is the enclosed electric charge, and *d* is the displacement.

When displacement of the test charge occurs from height y_A to y_B, the work done by the electric field on the charge is determined as follows:

$$W_{A \to B} = -\Delta U = -(U_B - U_A) = -(q_0 E y_B - q_0 E y_A) = q_0 E (y_A - y_B)$$

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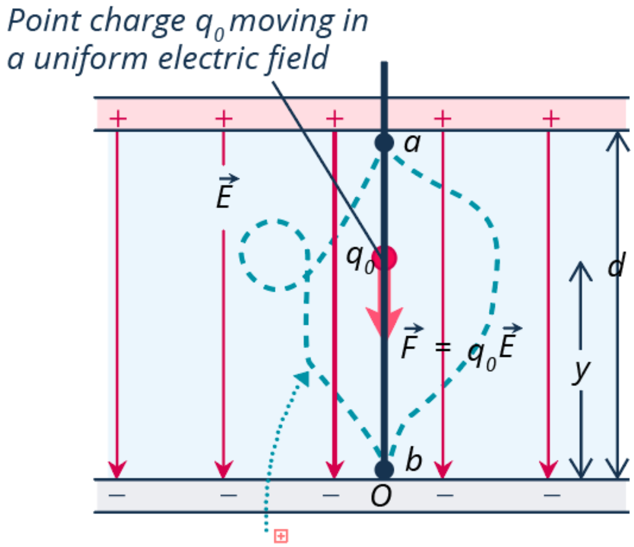

Fig. 3.1.3. The work done on point charge q_0 in a uniform electric field

Fig. 3.1.3, shows the relationship between the charged particle q_0, the direction of electric field E and the resulting potential energy. Two significant rules arise from this:

1. The potential energy U increases if the charge q_0 moves in the direction opposite of the electric force.
2. The potential energy U decreases if the charge q_0 moves in the direction similar to the electric force.

How do we determine the potential energy of a charged particle in a uniform electric field?

Electric Potential Energy of Two Charges

In some cases, the potential energy U becomes a "shared property" of two charges (q and q_0), as shown in **Fig. 3.1.4**. Here, a charge exerts force on another charge and potential energy results from the interaction of charges.

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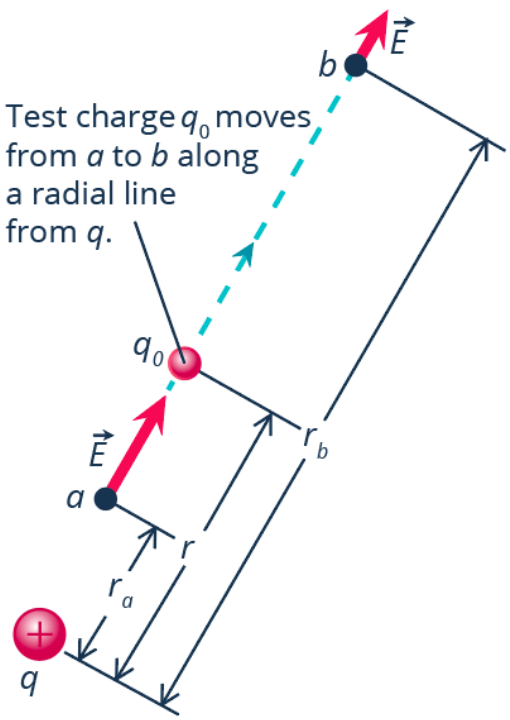

Fig. 3.1.4. Charge q_0 extending radially from charge q

For two charges q and q_0 separated by a certain distance r, the electric potential energy arising from their interaction is mathematically expressed as follows:

$$U_E = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r}$$

where $1/4\pi\varepsilon_0$ is also denoted as k or Coulomb's Constant, equal to 9.0×10^9 Nm^2/C^2, q and q_0 are the electric charges, and r is the separation distance between q and q_0.

How do we determine the potential energy of two point charges?

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Tips

Keep in mind that the potential energy *U* increases if the charge q_0 moves in the direction opposite of the electric force and it decreases if the charge q_0 moves in the direction opposite of the electric force.

Let's Practice!

Example 1 EASY

One electron and one proton of a hydrogen atom are at a distance of 4.1 $\times$ 10^{-11} m apart. Determine the potential energy of the electron in the proton's field.

Solution

Step 1: Identify what is required in the problem.

You are asked to calculate the potential energy U_E of the electron in the proton's field.

Step 2: Identify the given in the problem.

The distance r is given.

The charge of a proton is equivalent to 1.6 $\times$ 10^{-19} C and the charge of an electron is -1.6 $\times$ 10^{-19} C.

$$\mathrm{r} = 4.1 \times 10^{-11}\mathrm{m}$$

$$\mathrm{q} = 1.6 \times 10^{-19}\mathrm{C}$$

$$\mathrm{q_0} = -1.6 \times 10^{-19}\mathrm{C}$$

Step 3: Write the working equation.

$$\mathrm{U_E} = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{\mathrm{r}}$$

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Step 4: Substitute the given values.

$$U_E = 9.0\times10^9\frac{Nm^2}{C^2}\left(\frac{1.6\times10^{-19}C(-1.6\times10^{-19}C)}{4.1\times10^{-11}m}\right)$$

Step 5: Find the answer.

$$U_E = -5.61\times10^{-18}Nm \text{ or } J$$

Thus, the **potential energy of the two charges** is equal to **–5.61 × 10^{-18} Nm or J**. The U_E is negative since the charges are opposite, hence, attracting each other.

> **1 Try It!** EASY
>
> Two protons are at a distance of 2 × 10^{-15} m from each other. What is the potential energy of the two protons?

Example 2 AVERAGE

Two point charges (C and D) are separated from each other by a distance of 3.51 m. Calculate the magnitude of the electric charge of particle D if particle C has 9.3 μC and their potential energy results to 781 J.

Solution

Step 1: Identify what is required in the problem.
You are asked to calculate the magnitude of the electric charge of particle D.

Step 2: Identify the given in the problem.
The distance between the two particles 3.51 m, the charge of particle C 9.3 μC, and the potential energy 781 J are given.

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Step 3: Write the working equation.

$$U_E = k\frac{q_C q_D}{r}$$

Derive the formula for q_C*:*

$$q_D = \frac{U_E r}{k q_C}$$

Step 4: Substitute the given values.

$$q_D = \frac{781\,\mathrm{J}\,(3.51\,\mathrm{m})}{9.0 \times 10^9\,\frac{\mathrm{Nm2}}{\mathrm{C}^2}\,(9.3 \times 10^{-6}\,\mathrm{C})}$$

Step 5: Find the answer.

$$q_D = 0.0328\,\mathrm{C}$$

Thus, the **magnitude of the electric charge** of particle D is **0.0328 C**.

2 Try It! AVERAGE

Charges E and F are situated 4.44 m apart. What is the magnitude of the electric charge of particle F if particle E has 11.2 μC and their U_E is 914 J?

Example 3 DIFFICULT

The electric potential energy U_E experienced by an electron situated from a proton with 2e at a distance of 4.53 pm is 650 MJ. Find the magnitude of the electron.

Solution

Step 1: Identify what is required in the problem.
You are asked to calculate the magnitude of the electron.

Step 2: Identify the given in the problem.
The electric potential energy (650 MJ) and the distance between the two charges (4.53 pm) are given.

The charge of two protons (2e) is equivalent to $2(1.6\times10^{-19}\mathrm{C})$.

Step 3: Write the working equation.

$$q_e = \frac{\mathrm{U_E\,r}}{\mathrm{k}\,q_p}$$

Step 4: Substitute the given values.

$$q_e = \frac{650\,\mathrm{MJ}(4.53\,\mathrm{pm})}{k(2(1.6\times10^{-19}\mathrm{C}))}$$

Convert the necessary values to their corresponding SI units.

$$q_e = \frac{(6.50\times10^{8}\,\mathrm{J})(4.53\times10^{-12}\,\mathrm{m})}{k(3.20\times10^{-19}\,\mathrm{C})}$$

Step 5: Find the answer.

$$q_e = -1.02\times10^{6}\,\mathrm{C} = -1.02\,\mathrm{MC}$$

Thus, the **magnitude of the electron** is equivalent to **–1.02 $\times$ 10^{6} C**.

3 Try It! DIFFICULT

A positively-charged particle with 4*e* is separated at a distance of 6.11 pm from a negatively-charged particle. Determine the magnitude of the electron if the electric potential energy of the two particles is 718.18 MJ.

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Key Points

- **Electric potential energy** refers to the energy needed to move a charge against an electric field. The work done by an electric force to move a charge is expressed in terms of potential energy U_E.
- The **law of conservation of mechanical energy**, when applied to electric charges, refers to the total kinetic and potential energies present in a system.
- In a uniform electric field, where the strength of the field is not altered, the work done by the electric field is determined by the product of the force magnitude and the displacement component in the downward direction of the force.
- When the potential energy is a shared property between two charges, a charge exerts force on another charge and potential energy results from the interaction of charges.

Key Formulas

Concept	Formula	Description
Work Done by the Electric Field	$W_{A\to B} = Fd = q_0Ed$ where • $W_{A\to B}$ is the net electric flux; • F is the force; • d is the displacement, and • q_0E is the enclosed electric charge.	Use this formula to solve the **work done by the electric field** when either the force and displacement or the enclosed electric charge and displacement are given.

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Electric Potential Energy in a Uniform Electric Field	$$U_E = q_0Ed$$ where • U_E is the electric potential energy; • d is the displacement, and • q_0E is the enclosed electric charge.	Use this formula to solve **the electric potential energy in a uniform electric field** when the displacement and the enclosed electric charge are given.
Electric Potential Energy of Two Charges	$$U_E = \frac{1}{4\pi\varepsilon_0}\frac{qq_0}{r}$$ where • U_E is the electric potential energy; • $1/4\pi\varepsilon_0$ or k is equivalent to $9.0 \times 10^9\ Nm^2/C^2$; (Coulomb's Constant) • q is the charge of first particle; • q_0 is the charge of second particle, and • r is the separation distance between q and q_0.	Use this formula to solve **the electric potential energy of two charges** when the magnitudes of the charges, and the separation distance between them are given.

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Check Your Understanding

A. Identify whether each of the following statements is true or false.

UNDERSTAND

_____________ 1. Kinetic energy refers to the stored energy of an object at rest.

_____________ 2. In a uniform electric field, the electric field lines are spaced at varying distances from each other.

_____________ 3. Electric force is not an example of a conservative force.

_____________ 4. The value of k, or Coulomb's constant, is 90×10^9 Nm^2/C^2.

_____________ 5. The standard unit for electric potential energy is the joule.

_____________ 6. Electric potential energy is the energy needed to move a charge against the electric field.

_____________ 7. The work done by the electric field is determined by the product of the force magnitude and the displacement component.

_____________ 8. The work done by the electric force to move an electric charge is expressed in terms of chemical energy.

_____________ 9. When the electric force is in a similar direction with the net displacement of the charge, the work is considered negative.

_____________ 10. The equation $K_A - U_A = K_B - U_B$ is also known as the law of the conservation of mechanical energy.

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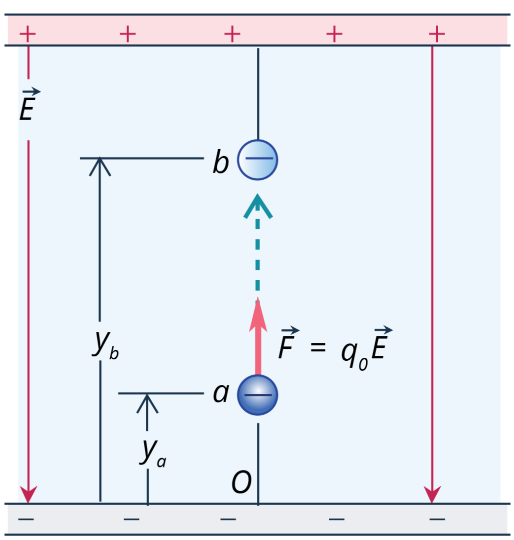
B. Answer the questions that follow. **APPLY**

1. One electron and one proton of a hydrogen atom are at a distance of 3.4×10^{-11} m apart. Determine the potential energy of the electron in the proton's field.
2. Two protons are at a distance of 6×10^{-15} m from each other. What is the potential energy of the two protons?
3. Two point charges are separated from each other by a distance of .003 cm. Charge C has a magnitude of 9 μC and charge D has a magnitude of -8 μC. What is the potential energy of the two charges?
4. Two electrons are at a distance of 3.8×10^{-15} m from each other. What is the potential energy of the two charges?
5. The U_E experienced by an electron situated from a proton with $2e$ at a distance of 1.78 pm is 430 MJ. What is the magnitude of the electron?
6. Charges F_1 and F_2 are at a distance of 1.99 m. What is the magnitude of the electric charge of F_2 if F_1 has a magnitude of 7.77 μC and the U_E of two particles is 432.21 J?

C. Analyze the illustrations below. Write A if the potential energy will increase, B if it will decrease, and C if it will remain unchanged. **ANALYZE**

1. ______________________



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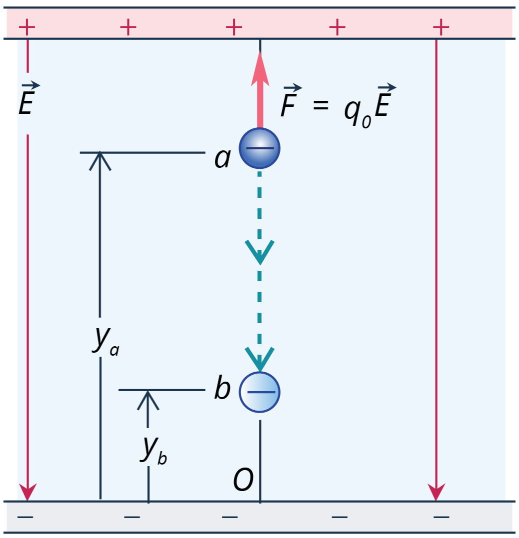
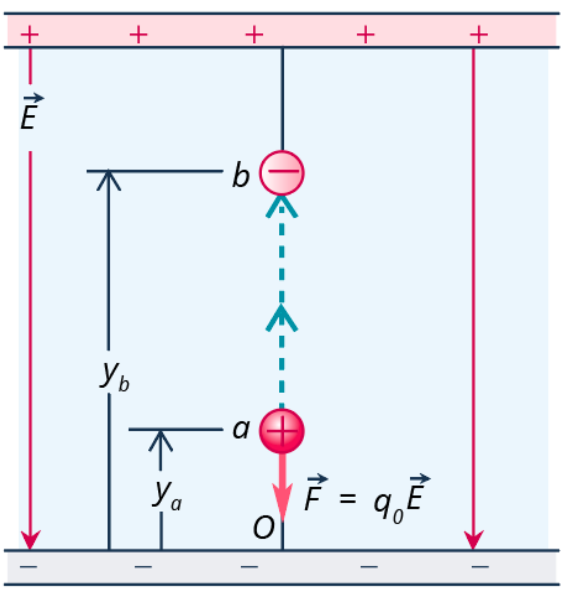
2. ____________________



3. ____________________



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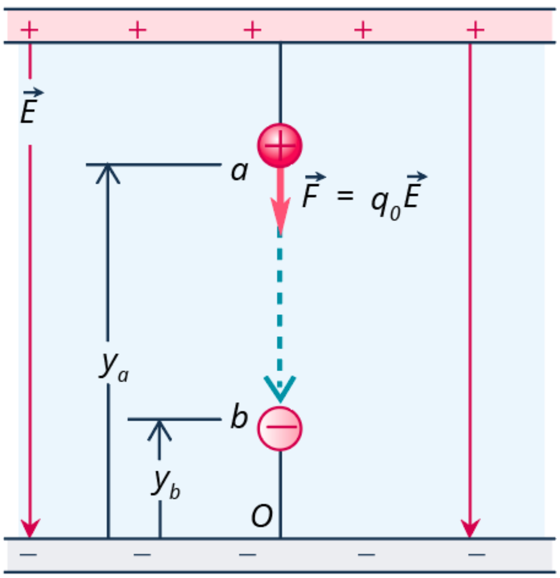
4. ______________________



Challenge Yourself

Answer the following questions.

EVALUATE

1. Determine the change in the distance between two equal charges if the electric potential energy U_E increases four times.
2. A positively charged particle, q_1 = +3.50 mC, is at rest at its point of origin. A second, negatively charged particle, q_2 = -6.70 m, is displaced from the point, x = 0.250m, y = 0, to the point, x = 0.340m, y = 0.370m. How much work is done by the electric force on q_2?
3. A particle q_1 is at rest at its point of origin. Another particle, q_2 is positioned at point a, yielding an electric potential energy of 6.3×10^{-8} J. As the second particle is moved to point b, the force on the charge produces work of -2.7×10^{-8} J. Determine the electric potential energy of the two charges when the second particle is at point b.
4. What must one do to double the electric potential energy of two charges?
5. Why is the electric potential energy of two similar charges positive at all times?

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Key to *Try It!*

1. 1.15×10^{-13} J
2. 0.0403 C
3. -7.62×10^{5} C